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STAT3375Q HW4

3.103 $\frac{\binom{6}{5}}{\binom{10}{5}} \approx 0.0238$

3.106 $p(\text{Defective}) = \frac{4}{10}$
 $E(Y) = np = \$50 \frac{4}{10} = \20

$Var(Y) = np(1 - p) = 20(.6) = \12

3.105 a Hypergeometric. We are selecting $n = 3$ randomly sampled objects from a population of size $N = 8$ with $r = 5$ Class 1 objects.

b $P(Y \geq 2) = P(Y = 2) + P(Y = 3)$
 $= \frac{\binom{5}{2}\binom{8-5}{3-2}}{\binom{8}{3}} + \frac{\binom{5}{3}\binom{8-5}{3-3}}{\binom{8}{3}} \approx .714$

c $E(Y) = n \frac{r}{N} = 3 \frac{5}{8} = 1.875$
 $Var(Y) = n \left(\frac{r}{N} \right) \left(\frac{N-r}{N} \right) \left(\frac{N-n}{N-1} \right) = 3 \frac{5}{8} \frac{8-5}{8} \frac{8-3}{8-1} \approx .502$

3.107 $N = 10, n = 4, r = 5$

a $P(Y = 4) = \frac{\binom{5}{4}\binom{10-5}{4-4}}{\binom{10}{4}} \approx .0238$

b $P(Y \leq 3) = 1 - P(Y = 4) \approx .976$

c $P(Y \geq 1) = 1 - P(Y = 0) \approx .976$

3.113 $N = 20, n = 6, r = 8$

$P(Y = 1) = \frac{\binom{8}{1}\binom{12}{5}}{\binom{20}{6}} \approx .163$, which is not negligible.

3.114 $E(Y) = 6 \frac{8}{20} = 2.4, Var(Y) = 6 \frac{8}{20} \frac{20-8}{20} \frac{20-6}{19} \approx 1.06$

3.118 $\frac{P(Y=4)}{P(Y=3)}$ $N = 52, n = 5, r = 4$

$\frac{\binom{4}{4}\binom{48}{1}}{\binom{52}{5}} \approx .0106$

3.121 a $P(Y = 4) = \frac{e^{-2}2^4}{4!} = \frac{2}{3}e^{-2}$

b $P(Y \geq 4) = 1 - P(Y < 4) = P(Y = 0) + P(Y = 1) + P(Y = 2) + P(Y = 3) \approx 1 - .1353 + .2707 + .2707 + .1804 \approx .143$

c $P(Y < 4) = 1 - P(Y \geq 4) = 1 - .143 = .857$

d $\frac{P(Y \geq 4)}{1 - P(Y < 2)} = \frac{.143}{1 - .1353 + .271} \approx .241$

3.123 $\frac{e^{-\lambda}\lambda^0}{0!} = \frac{e^{-\lambda}\lambda^1}{1!}$
 $\lambda = 1, p(2) \approx .184$

3.127 $\lambda = 4, P(Y \leq 4) \approx .0183 + .0733 + .1465 + .1954 + .1954 \approx .629$

3.139 $E(Y) = 50 - 2(2) - 2^2 = 42$

10 $P(a = 0) = \frac{e^{-1.3*4}(1.3*4)^0}{0!} \approx .0055$

$$P(b = 5) = \frac{\binom{10}{5} \binom{18-10}{5-5}}{\binom{18}{5}} \approx .0294$$

Therefore it is more likely that Martha spent the whole night with ghosts.